SPRINGBOARD DATA SCIENCE CAREER TRACK · CAPSTONE THREE

Consumer Credit Risk Modelling

Loan Default Prediction with Canadian Macroeconomic Stress Testing for Credit-Union Decision Support

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github.com/MIJUMBO/consumer-credit-risk-modelling

PROBLEM & CONTEXT

Three questions a credit committee must answer

Under IFRS 9, provisioning answers must be forward-looking and defensible — yet many mid-sized Canadian lenders still answer them with a single flat reserve rate.

1 WHO

Which borrowers are likely to default?

2 HOW MUCH

How much should we set aside against them today?

3 WHAT IF

How much more would a recession demand?

Why the flat-rate fallback fails

- Over-provisions low-risk borrowers; under-provisions high-risk ones
- Treats credit-impaired exposures no differently from performing loans
- Offers no mechanism to translate an economic scenario into a provision number

The stakes: for a billion-dollar consumer book, an imprecise ECL estimate can misallocate tens of millions in reserves — under-provisioning threatens capital adequacy in a downturn; over-provisioning ties up lendable capital.

FRAMING & SUCCESS CRITERIA

From a business question to a testable problem

PROBLEM STATEMENT

Can a leakage-free model, trained only on origination-time information, produce calibrated PDs accurate enough to (a) support IFRS 9 staging and provisioning, and (b) quantify how provisions change under adverse Canadian macro scenarios — giving a risk committee a defensible alternative to flat-rate reserving?

Revised criteria for success

- **Real, stable discrimination** — materially better than random on an out-of-time test, stable across vintages
- **Bankable calibration** — Brier beating the base rate; mean PD ≈ observed default rate
- **Complete IFRS 9 ECL** — staging, 12-month vs lifetime, baseline + stress table via one code path
- **Full explainability** — global and loan-level, for governance and adverse-action

The decision that reshaped "success"

The proposal targeted AUC ≥ 0.70. But excluding LendingClub's own risk-pricing fields *(grade, sub-grade, interest rate)* — which would teach the model to reproduce the lender's judgement rather than borrower risk — imposes an honest ceiling near **0.67 AUC.**

Value therefore migrated from *discrimination* to **calibration and provisioning** — exactly where a credit-risk practitioner locates it.

THE DATA & ARCHITECTURE

A three-layer design on public data

LAYER 1

1 Borrower risk

1,345,350 resolved LendingClub consumer loans (2007–2018). Application, credit profile, and final outcome.

LAYER 2

2 Macro conditions

Five-indicator Canadian panel (Bank of Canada + Statistics Canada), 2000–2026, 318 monthly rows.

LAYER 3

3 Portfolio ECL

Integrates the two at portfolio level through IFRS 9 stress scenarios and Expected Credit Loss.

Architectural rule: the macro panel is never joined to loan-level data. US borrower records carry no meaningful row-level link to Canadian indicators — the macro layer informs scenario design only. This standard IFRS 9 separation is what makes the ***methodology*** — not the dataset — the transferable contribution.

Sources: LendingClub accepted loans (Kaggle) · Bank of Canada Valet API · Statistics Canada (unemployment, CPI, debt-service ratio, insolvencies)

DATA WRANGLING

Building a clean, leakage-free training base

2.26M
raw loans

1.35M
resolved outcomes retained

19.96%
overall default rate (1:4)

32
temporally stable features

Leakage control: two distinct guardrails

DROPPED ENTIRELY — 22 cols

Post-outcome fields — repayment totals, recoveries, last-payment dates, post-origination FICO pulls. Available only because the loan already resolved; training on them inflates performance to meaningless levels.

RETAINED FOR EDA, NEVER MODELLED

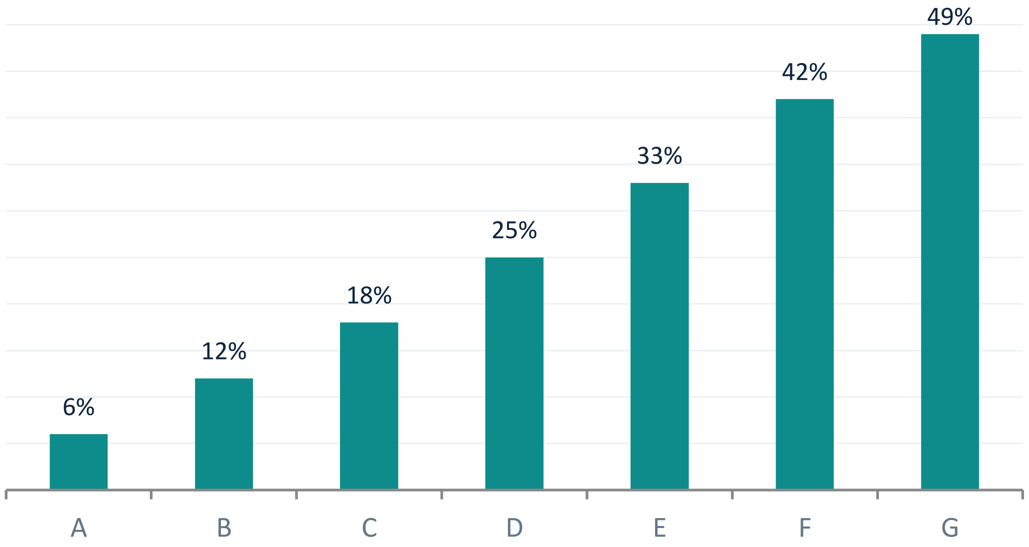
grade, sub-grade, interest rate are origination-time information, but modelling them predicts the lender's own pre-computed judgement rather than borrower risk. Kept as EDA benchmarks only.

Cross-vintage audit: LendingClub's schema evolved 2007–2018; each column was classified by missingness per vintage year, keeping only consistently available features.
Sampling: stratified on issue-year × default-flag — 100k dev sample for iteration, 500k production sample for final benchmarks.

EXPLORATORY ANALYSIS

Borrower-level findings

Default rate climbs monotonically with lender grade



Grade is predictive — and precisely why it must be excluded to avoid circularity.

Loan purpose separates risk

Small business (27.7%), renewable energy (25.3%) default most; car (16.0%) and wedding (16.2%) least — motivating a 3-tier purpose-risk feature.

Vintage matters by year, not quarter

Default climbs ~13% (2008–09) to ~23% (2016–17) as standards loosened; calendar quarter spreads only 1.3pp.

All correlations are weak

FICO strongest protective (−0.129), DTI strongest risk (+0.093). Default is non-linear and multi-factor — the case for tree models.

EXPLORATORY ANALYSIS

Macroeconomic findings anchor the stress design

Canadian unemployment, 2000–2026

GFC = the stress anchor

+2.7pp

Largest sustained unemployment rise in the 26-year panel (6.1% → 8.8%, 2008–09). The COVID spike (+8.7pp) was larger but artificial and rapidly reversed — a poor template for sustained credit stress.

Why elasticity is an assumption

Insolvencies co-move with unemployment in normal times and through the GFC — but decoupled during COVID, when transfers and payment deferrals severed the usual link. The PD–unemployment elasticity is therefore treated as a scenario assumption, not an estimated law.

FEATURE ENGINEERING

Fifteen origination-time predictors, one source of truth

01

Affordability

loan-to-income · DTI · loan amount · log income

02

Credit quality

FICO (range midpoint) · revolving util. · open-acc band

03

Derogatory flags

delinquency flag · public-record flag

04

Employment

employment years · short-tenure flag (<1 yr)

05

Loan & vintage context

purpose-risk tier · home ownership · vintage year & quarter

Design lesson

DTI, FICO, and revolving utilisation enter as raw continuous values rather than coarse bands — binning discarded signal and measurably lowered performance. Tree models choose their own split points. Encoding is ordinal (fitted on train only), not one-hot, since the champions are gradient-boosted trees.

Temporal split — train on history, test on the future

Split	Vintages	Default rate
Train	≤ 2015	18.43%
Validation	2016	23.29%
Test (out-of-time)	2017–18	21.29%

MODELLING

Three models, one evaluation framework

Out-of-time validation performance

Model	AUC	KS	Gini	Brier
XGBoost (champion)	**0.6684**	**0.2472**	**0.3367**	**0.2316**
LightGBM	0.6684	0.2471	0.3369	0.2320
Logistic Regression	0.6616	0.2362	0.3232	0.2368

Base-rate Brier reference: 0.1786 at a 23.3% default rate.

A feature-set ceiling, not a tuning limit

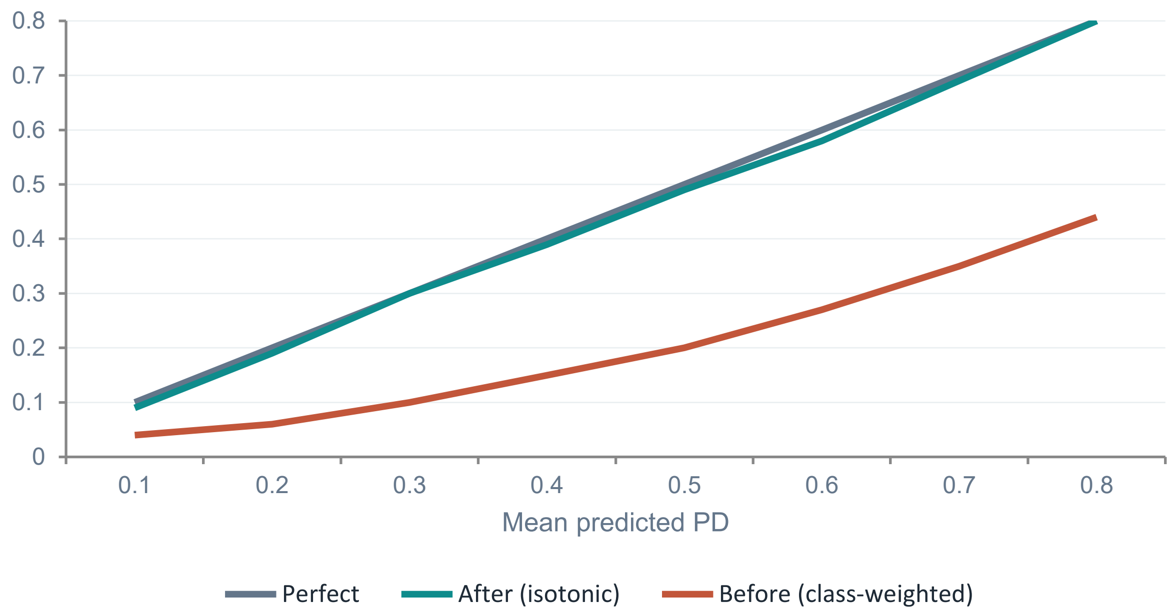
Two boosters tie to four decimals from entirely different configurations *(XGBoost: 230 depth-4 trees; LightGBM: 591 trees at 15 leaves)*. Independent searches converging on the same score is evidence of a stable ceiling set by the leakage-free feature set — the intended price of excluding the lender's own risk pricing.

Champion = XGBoost on parsimony (230 vs 591 trees — lighter to deploy and monitor) and a marginally better Brier; Logistic Regression is kept as an interpretable benchmark. **Dollar-weighted operating threshold (0.490)** catches 69% of future defaulters — diagnostic only; the IFRS 9 framework consumes continuous PDs.

THE KEY RESULT

Calibration — turning scores into valid PDs

Reliability curve — before vs after isotonic calibration



Class weighting inflates raw probabilities

The champion's raw Brier (0.2254) was worse than a constant base-rate guess — loans scored ~55% PD defaulted only ~26% of the time. Used directly in ECL, these would overstate provisions materially.

~~0.2254~~ → **0.1576**

Post-calibration Brier beats the 0.1676 base rate — the single result that licenses multiplying PDs by dollars.

Isotonic regression is monotonic — AUC, KS, and Gini are untouched; only the probability scale is corrected.

VALIDATION

Discrimination is real, stable, and honestly bounded

0.6726	0.6725	0.6668
Held-out test AUC	**2017 vintage AUC**	**2018 vintage AUC**
out-of-time 2017–18 vintages	*no temporal degradation*	*within sampling noise*

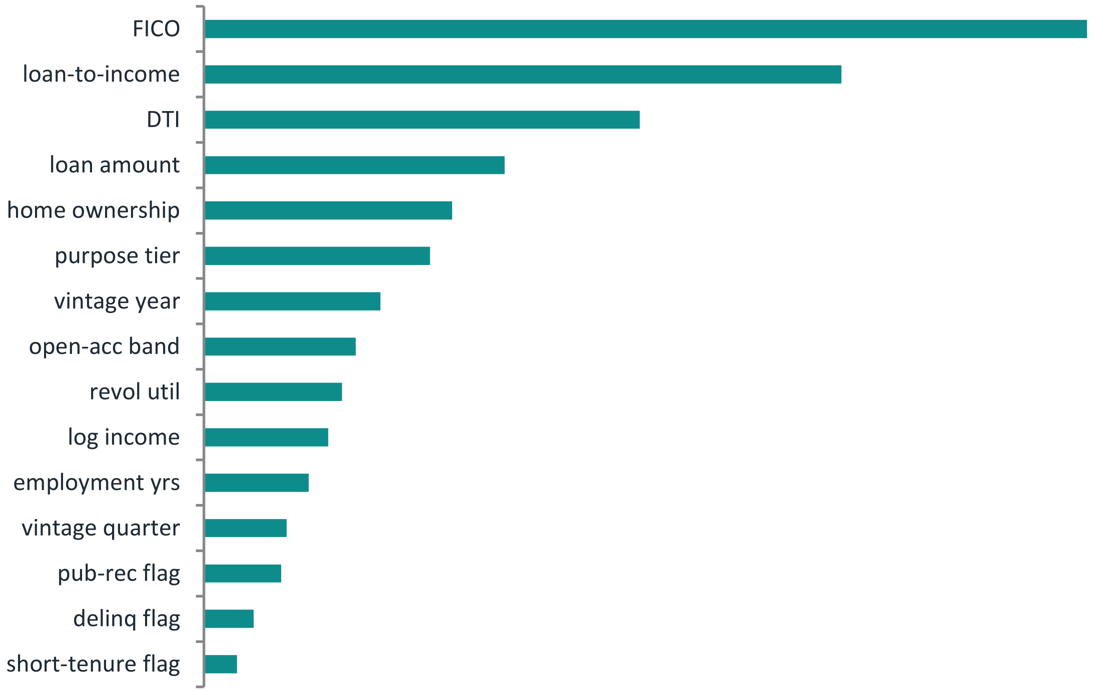
The ~0.67 ceiling belongs to the feature set, not the algorithm

Two independent hyperparameter searches converging on identical validation AUC from different architectures shows the plateau is a property of the leakage-free origination feature set — the deliberate, honest price of excluding the lender's own risk pricing. The model generalises across loan cohorts rather than memorising one economic period.

EXPLAINABILITY

SHAP — the model relies on economically meaningful drivers

Global feature importance (mean |SHAP|)



GLOBALLY

Credit quality and affordability dominate — FICO, loan-to-income, DTI — with engineered features (home ownership, purpose tier, vintage) contributing meaningfully.

DIRECTIONALLY

Higher leverage and indebtedness push risk up; stronger FICO, income, and tenure pull it down — consistent with credit theory.

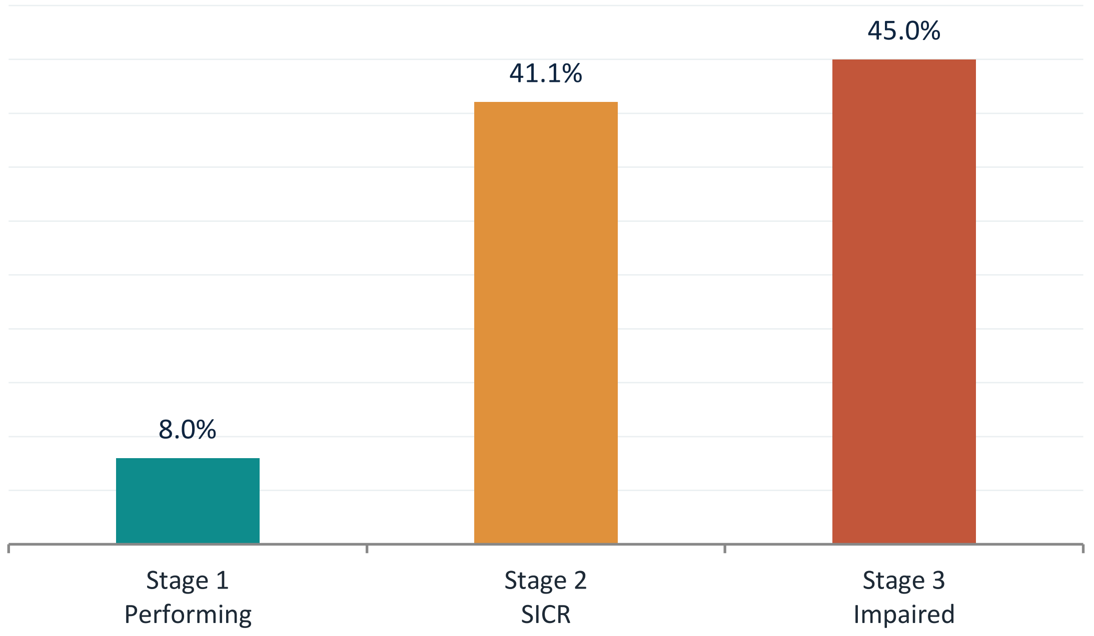
LOCALLY

A waterfall plot decomposes any single decision into exact feature contributions — the auditable rationale an adverse-action notice requires.

IFRS 9 ECL

Baseline allowance — and the value of the model in dollars

Coverage rises sharply with credit deterioration



Baseline allowance

$56.8M 23.5% coverage

on the $242.1M held-out test portfolio

vs. a naive flat-rate reserve

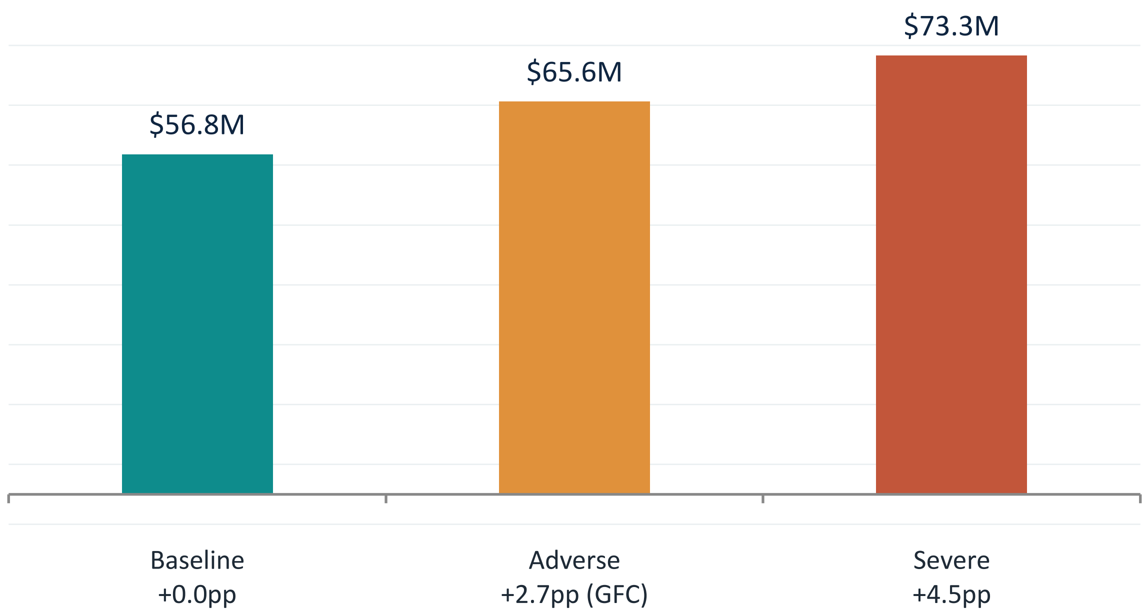
$23.2M held — **59% less**

It treats impaired loans like performing ones and ignores lifetime risk concentrated in SICR.

The model's value is risk differentiation — directing provisions to the loans most likely to generate losses while avoiding unnecessary reserving on low-risk exposures.

Provisions under scenarios anchored in Canadian history

IFRS 9 Expected Credit Loss by scenario



ADVERSE (GFC REPLAY)

+15.5%

+$8.8M of required provisions

SEVERELY ADVERSE

+29.0%

+$16.5M; beyond-sample at ~1.7× the GFC

Two channels lift provisions at once

- Every PD lifts (+10% relative per +1pp unemployment)
- Loans migrate Stage 1 → Stage 2 across the fixed SICR line, switching from 12-month to lifetime ECL
- Stage 2 population roughly doubles: 2,514 → 5,022
- Stage 3 stays fixed at 3,571 — impairment is realised, not scenario-dependent

Anchored, not round: Adverse replays the GFC exactly; Severe is deliberately beyond-sample.

RECOMMENDATIONS

What the client should do

1 Adopt model-based, staged provisioning

Replace flat-rate reserving — price risk where it sits (8.0% / 41.1% / 45.0% coverage across stages). Begin with a parallel run, reconciling allowances quarterly before cut-over.

2 Pre-position a stress buffer; monitor SICR migration

Plan for +$8.8M (GFC) to +$16.5M (severe) of provisions on a $242M book. Track the count of loans approaching the SICR threshold monthly — it moves before losses do.

3 Recalibrate assumptions to internal data

LGD (45%), the lifetime multiplier (2.5×), and the PD–unemployment elasticity (10%/pp) are industry placeholders. Replace with figures from the institution's own loss history; re-fit the calibrator on the live book.

PRACTICAL CONSIDERATIONS

Suggestions for improvement & further research

Lift the feature-set ceiling

Credit-bureau trade-lines, behavioural or alternative data attack the binding ~0.67 constraint directly — a data problem, not a model problem.

Model the lifetime PD term structure

Replace the 2.5× approximation with survival analysis (Cox / discrete-time hazard) for exact, month-by-month lifetime ECL.

Estimate the macro elasticity

Regress vintage-level default rates on the Canadian panel (handling the COVID decoupling), allowing non-linear, regime-dependent responses — home of the originally scoped SARIMAX / Prophet work.

Production hardening

PSI monitoring, scheduled recalibration, champion–challenger governance, fairness testing, and a lightweight interactive scoring / stress app for risk committees.

Canadian-native validation

Partner with a credit union to re-estimate on domestic loan-performance data, closing the US-train / Canada-deploy geographic gap.

THE CENTRAL LESSON

The model is not the product — the provision number is.

A modest, honestly bounded AUC, once calibrated, allocates reserves to the loans that actually generate losses, flags significant increases in credit risk before they become defaults, and quantifies the capital a downturn would require — the three things a flat rate cannot do.

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